

Course Code : MCS-230
Course Title : Digital Image Processing and Computer Vision
Assignment Number : MCAOL(IV)/230/Assign/2025-26
Maximum Marks : 100
Weightage : 30%
Last Dates for Submission : 31st October, 2025 (For July, 2025 Session)
15th April, 2026 (For January, 2026 Session)

This assignment has sixteen questions of 5 Marks each; answer all questions. Rest 20 marks are for viva voce. You may use illustrations and diagrams to enhance the explanations. Please go through the guidelines regarding assignments given in the Programme Guide for the format of the presentation.

Q1: Explain the Optical, Analog, and Digital Image processing. **(5 Marks)**

Q2: (i) What should be the physical size of a 2D image of a document with dimensions is 3200x2400, when scanned at 400dpi. Here dpi stands for dots per inch. **(2x2.5=5 Marks)**

(ii) If the physical size of a medical image is 4×4 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better-quality image? Will an image of size 512×512 be enough?

Q3: (i) What is the storage requirement for a 2048 x 2048, 24-bit color image? **(2+3=5 Marks)**

(ii) Explain Intensity, Contrast, Brightness, Noise and Resolution with respect to Images.

Q4: (i) Check whether the matrix $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$ is unary or not? **(2.5+2.5=Marks)**

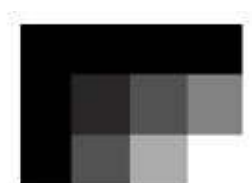
(ii) Perform the transformation $g_1(v) = 3v$ on the image $f(x,y) = \begin{bmatrix} -2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

Q5: Apply the given 3x3 mask w on the following image F (x, y) to generate the new image.

(5 Marks)

$$F(x, y) = \begin{bmatrix} 10 & 3 & 1 & 2 & 20 \\ 2 & 4 & 7 & 2 & 2 \\ 5 & 2 & 2 & 4 & 6 \\ 1 & 4 & 5 & 3 & 7 \\ 6 & 1 & 6 & 5 & 8 \end{bmatrix}, w(i, j) = \frac{1}{9} * \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q6: What is a histogram? Find the histogram of the image given below: **(5 Marks)**



0	0	0	0
0	1	2	3
0	2	4	6

- Q7:** Why is DCT important for image compression? Explain with the help of an example. **(5 Marks)**
- Q8:** What do you understand by Image enhancement? Explain the techniques of image enhancement with a suitable example. Also, discuss the advantages of image enhancement. **(5 Marks)**
- Q9:** Explain the following Smoothing Filter(s): **(5 Marks)**
- (i) Ideal Low Pass Filters (ILPF) (ii) Butterworth Low Pass Filters (BLPF)
- (iii) Gaussian Low Pass filters (GLPF)
- Q10:** What do you understand by feature extraction? What are its applications? Also discuss a few traditional methods of feature extraction. **(5 Marks)**
- Q11:** Explain image degradation and its types. **(5 Marks)**
- Q12:** Transform the RGB cube into its CMY cube. Label all the vertices. Also, interpret the colours at the edges with respect to saturation. **(5 Marks)**
- Q13:** Do you mean by Camera Calibration? Explain how intrinsic and extrinsic parameters of a camera are estimated? **(5 Marks)**
- Q14:** Explain Bayesian Classification with the help of a suitable example. **(5 Marks)**
- Q15:** Explain K-means clustering methods with the help of a suitable example. Also, discuss the advantages and disadvantages of k -K-means clustering methods. **(5 Marks)**
- Q16:** Perform partitional clustering using Froggy's method for the data given in the table below with k-2 (two clusters). Use first two sample points (3,3) and (6,8) as seed points. **(5 Marks)**

S. No.	X	Y
1	3	3
2	6	7
3	8	10
4	4	5
5	6	6
6	12	10
7	15	14
8	18	16